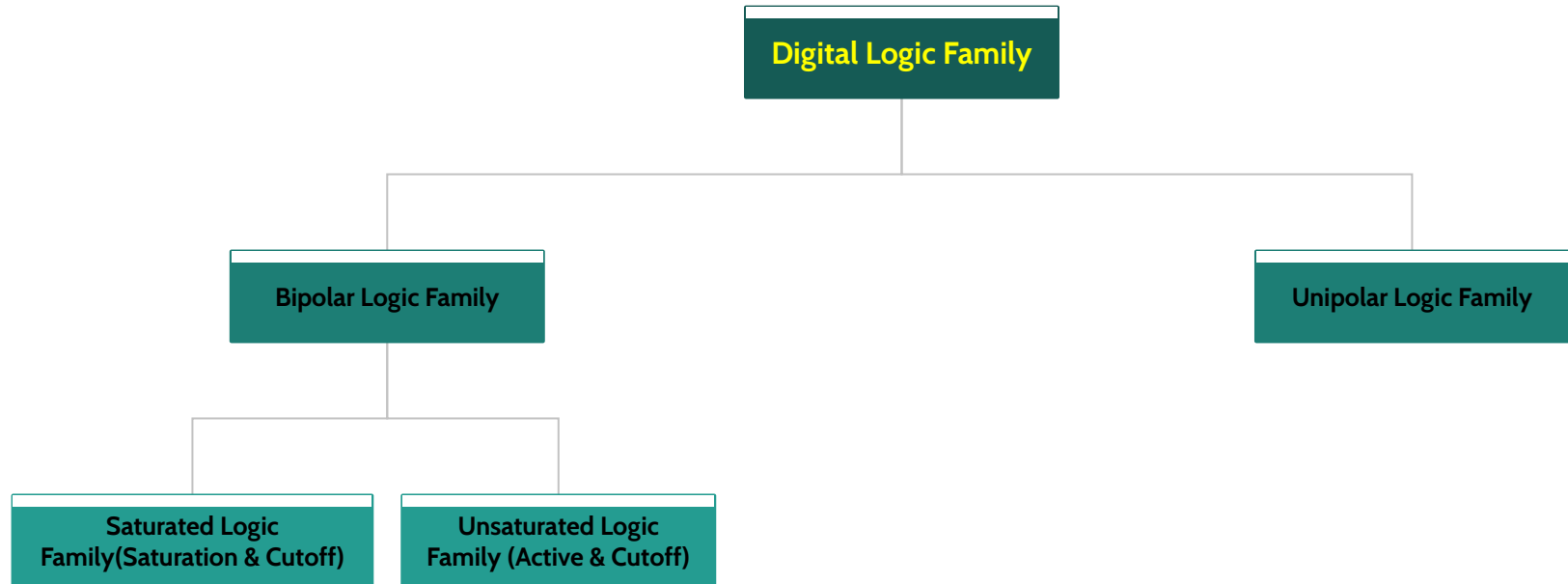


CSE350: Digital Electronics & Pulse Techniques

Lecture 03: Introduction to Digital
Logic Families

Digital Logic Family: *Different groups of electronic circuits that can be used to construct digital logic gates.* The logic gates we have used till now, have been abstract models of the logical function they execute. Digital logic family is the study of their actual circuit representations.



Saturated Logic Family: Uses *Saturation & Cutoff* region of a BJT for implementation of logic states.

1. Resistor Transistor Logic (RTL)
2. Diode Transistor Logic (DTL)
3. Direct Coupled Transistor Logic (DCTL)
4. Integrated Injection Logic (IIL)
5. High Threshold Logic (HTL)
6. Transistor Transistor Logic (TTL)

Unsaturated Logic Family: Uses *Active & Cutoff* region of a BJT for implementation of logic states.

1. Schottky TTL
2. Emitter Coupled Logic (ECL)

However, there is no restriction that Saturated & Unsaturated logic families cannot operate in Active & Saturated regions respectively.

Unipolar Logic Family: *JFET, MOSFET etc. single charge carrier based transistor logic families.*

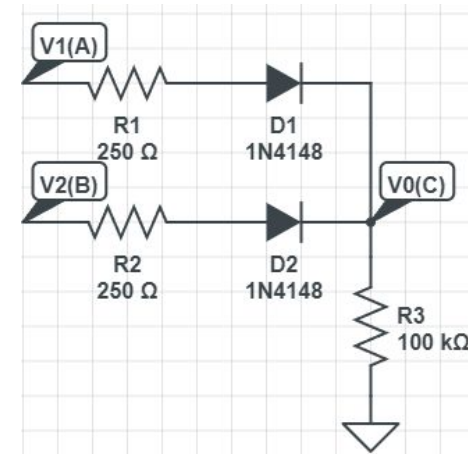
1. PMOS Logic Family
2. NMOS Logic Family
3. CMOS Logic Family

The highlighted logic families are included in our course.

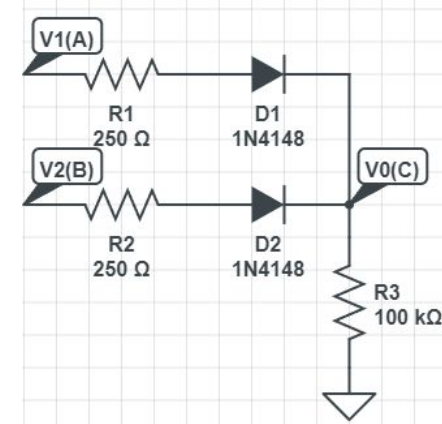
Diode Logic: The fact that a diode has two opposing biasing & models for those biasings, enables it to be used in building logic circuits. Let's look at some basic logic gates using diodes.

Ex-1: OR Gate:

We assume 5 V for logic '1' & 0 V for logic '0'. The truth table for a two input or gate is as follows:

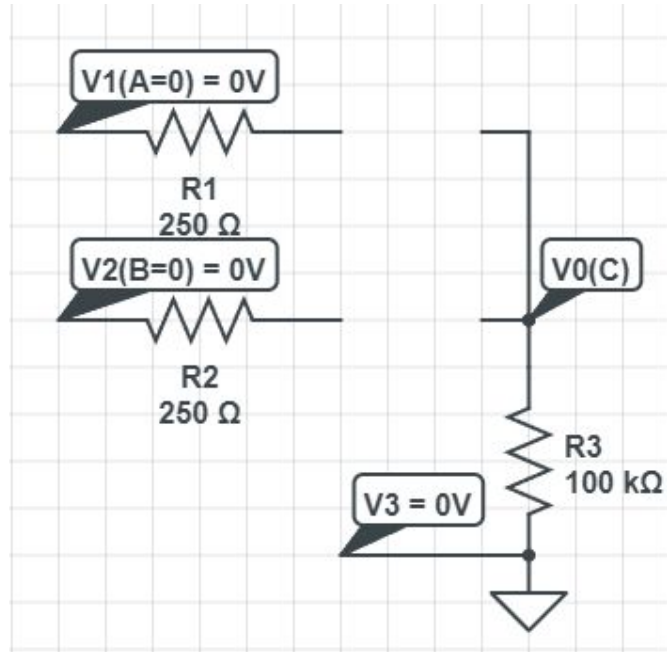


A	B	C	V1	V2	V0
0	0	0	0	0	0
0	1	1	0	5	4.28
1	0	1	5	0	4.28
1	1	1	5	5	4.29



For verifying each case, we will have to perform DC analysis on the circuit. For that, we will replace the diodes which will be on using **constant voltage drop model (0.7 V)**. To ease the process of determining which diodes will be on, we may momentarily assume them as ideal diodes & anticipate the voltage at both their terminals.

Case (0,0): When both inputs A & B are 0, both the diodes are off, hence they are simply replaced by open circuit. As we can see, the output is simply the ground voltage, 0 V. Thus we get logic '0'.



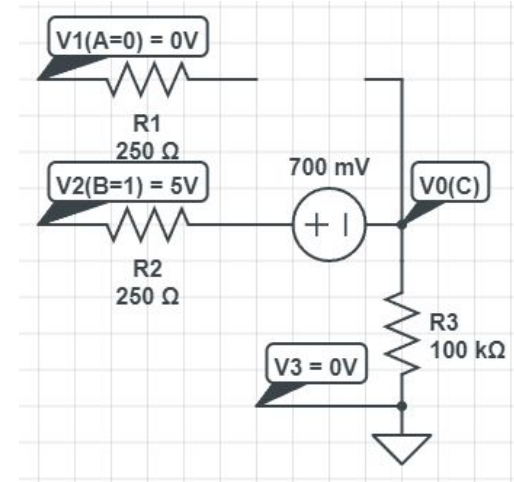
Case(0,1|1,0): Since the two diode branches are same, it is sufficient to perform analysis of either case (0,1) or case (1,0). Let us analyse (0,1). This means $V_1 = 0V$ & $V_2 = 5V$. Thus, the diode of input A's branch will be off & that of B will be on. So, we get,

Applying KCL,

$$\frac{V_0 - 0}{100k} + \frac{V_0 + 0.7 - 5}{0.25k} = 0$$

Solving,

$$V_0 = 4.28V$$



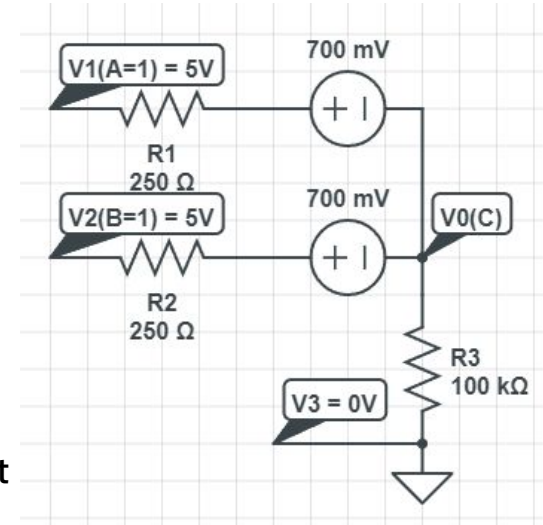
Case(1,1): Both diodes will be on this time. The resulting circuit is given below:

Applying KCL,

$$\frac{V_0 - 0}{100k} + 2 \frac{V_0 + 0.7 - 5}{0.25k} = 0$$

Solving, $V_0 = 4.29V$

As we can see, there are **2 different voltages for different cases in output voltage**. Since all these cases correspond to logic '1', we must select **the lowest one as the logic '1'** for proper interpretation at the output.



Power Dissipation: Suppose, we want to find the maximum power dissipated in the circuit among the four cases. To do this, we use the simple concept of power dissipation in an element, **Power dissipation = Voltage difference * Current.**

We may as well find it for all four cases to be exactly sure which one dissipates maximum power, but in our example through a bit of brainstorming, we can easily realize that the maximum power dissipation occurs in case (1,1), when both diodes are on.

Thus maximum dissipated power =

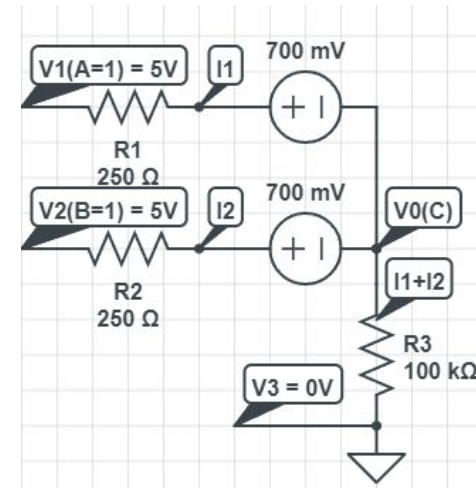
$$(V_1 - V_0) * I_1 + (V_2 - V_0) * I_2 + (V_0 - 0) * (I_1 + I_2)$$

This can actually be calculated in a short manner as follows,

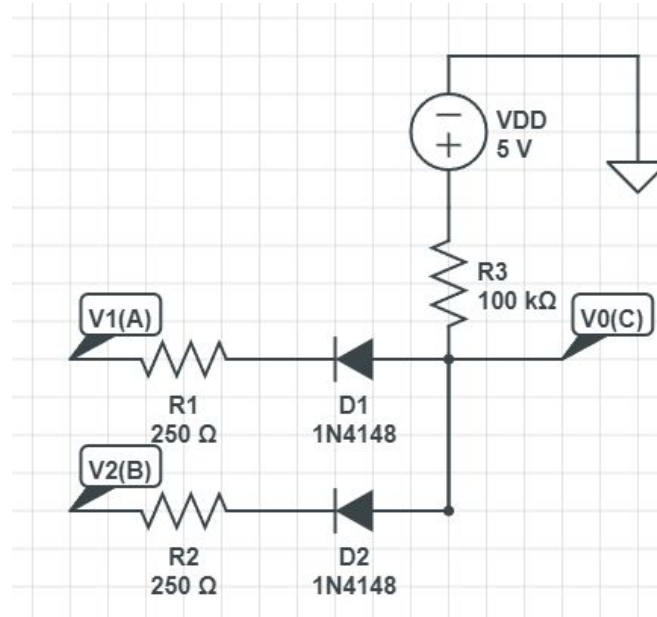
$$(V_1 - 0) * (I_1 + I_2) = (5 - 0) * 2 * I_1$$

$$\text{Now, } I_1 = \frac{V_1 - V_0 - 0.7}{0.25k} = \frac{5 - 4.29 - 0.7}{0.25k} = 0.04mA$$

$$\text{So, } P_{max} = 5 * 2 * 0.04 = 0.4mW$$



Practice Problem: Verify the following circuit as a logical AND gate in the similar manner as the previous example on OR gate.

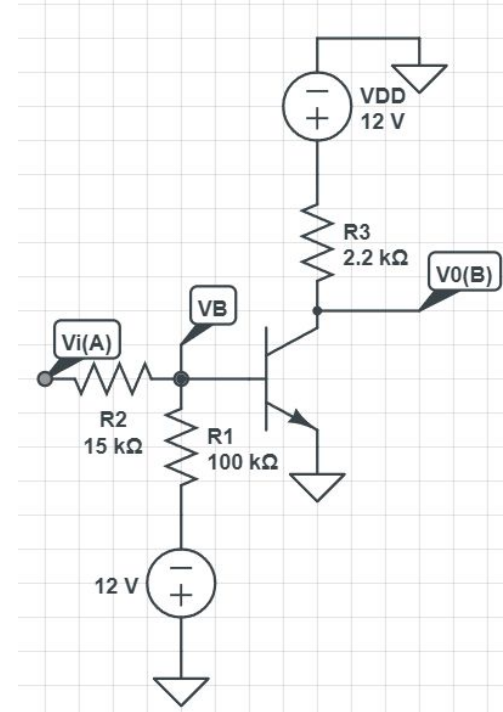


Resistor Transistor Logic (RTL)

RTL Inverter: Let's verify that the given circuit acts as an inverter.

A	B	V_i	V_O
0	1	0	12
1	0	12	0.2

For DC analysis of the circuit, we need to know the operating mode of the BJT in each logic case. We can see, that for output to be high, when input is low, requires the BJT to be in cutoff, so that it is an open circuit. So, we will assume it to be in cutoff & later verify it through analysis using the check condition for cutoff.



In cutoff, $I_C = I_B = I_E = 0$

$$\text{KCL at base, } \frac{V_B - (-12)}{100k} + \frac{V_B - 0}{15k} = 0$$

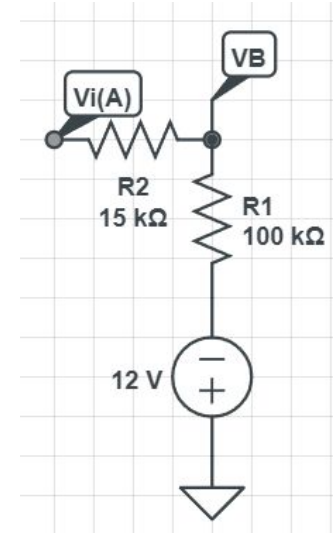
$$V_B = -1.565V \quad V_E = 0V$$

So, $V_{\lambda} = V_{BE} = -1.565V < 0.5V$ (Checking Condition)

We can thus conclude that, our assumption was correct. Now,

$$I_C = \frac{12 - V_0}{2.2k} = 0$$

$$V_0 = 12V$$



Now, let us do the same for the second case. For high input, the output must be low. Since, RTL is a saturated logic family, we may as well assume that the operating region here will be **Saturation** mode. The consequent DC analysis is shown next.

We recall, that in saturation mode, $V_{CE} = 0.2V$ & $V_B = 0.8V$

Now, $V_0 = V_C = V_{CE}$ (Since $V_E = 0V$)

So,
$$I_C = \frac{12 - 0.2}{2.2k} = 5.36mA$$

$$I_1 = \frac{12 - 0.8}{15k} = 0.746mA$$

$$I_2 = \frac{0.8 - (-12)}{100k} = 0.128mA$$

KCL at base gives, $I_B = I_1 - I_2 = 0.619mA$

Thus it satisfies the checking condition,

$$\beta_{forced} = \frac{I_C}{I_B} = 8.659 < 30 = \beta_F$$

So, our assumption was valid. The output voltage 0.2 V represents logic '0' or low.

